INCH-POUND

MIL-DTL-16377/73 (SH)
27 September 2004

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, EMERGENCY LIGHTING 20 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 331.1EM, 331.2EM, 347.2EM AND 347.3EM

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this document and the latest issue of MIL-DTL-16377.

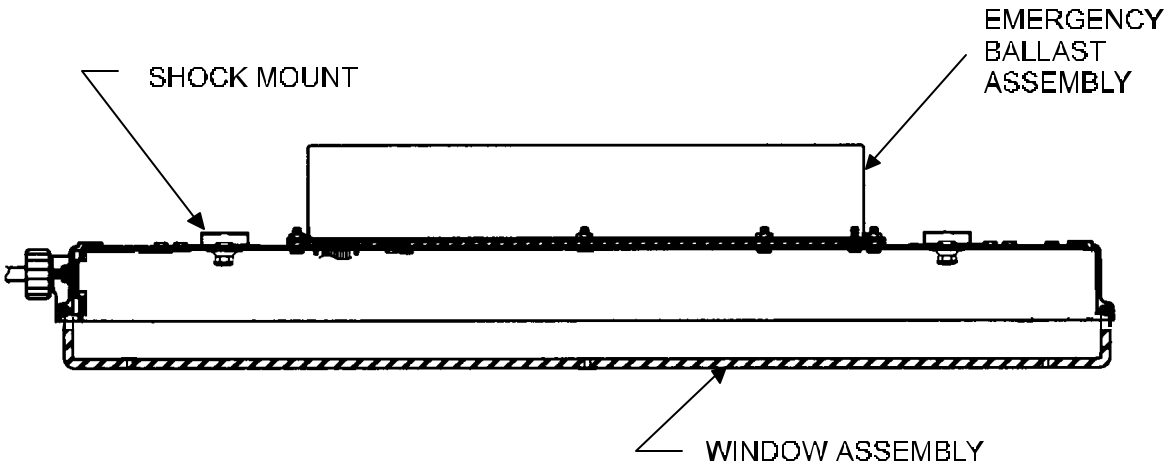
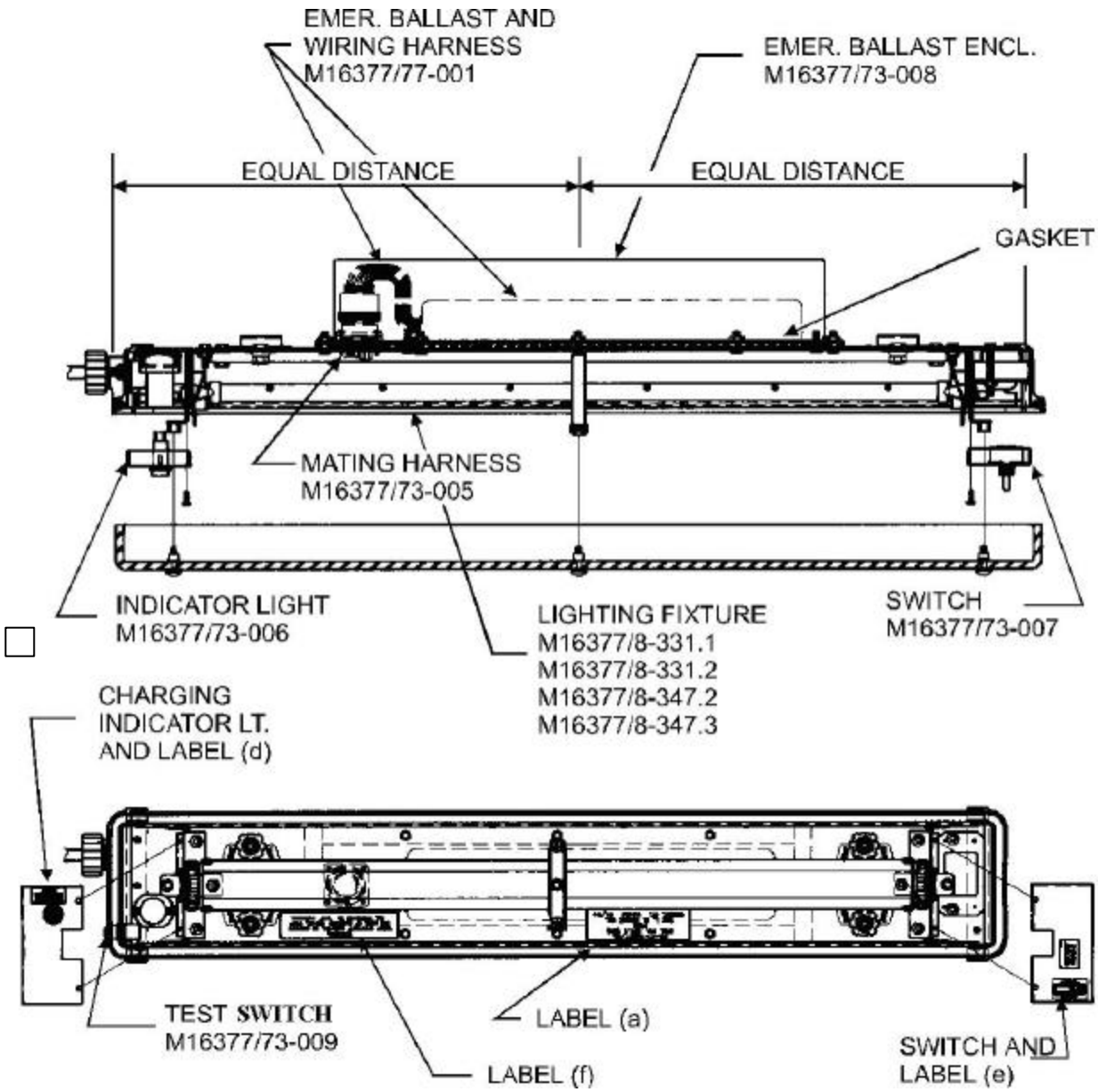


FIGURE 1. Typical outline of symbols 331.1EM, 331.2EM, 347.2EM and 347.3EM.

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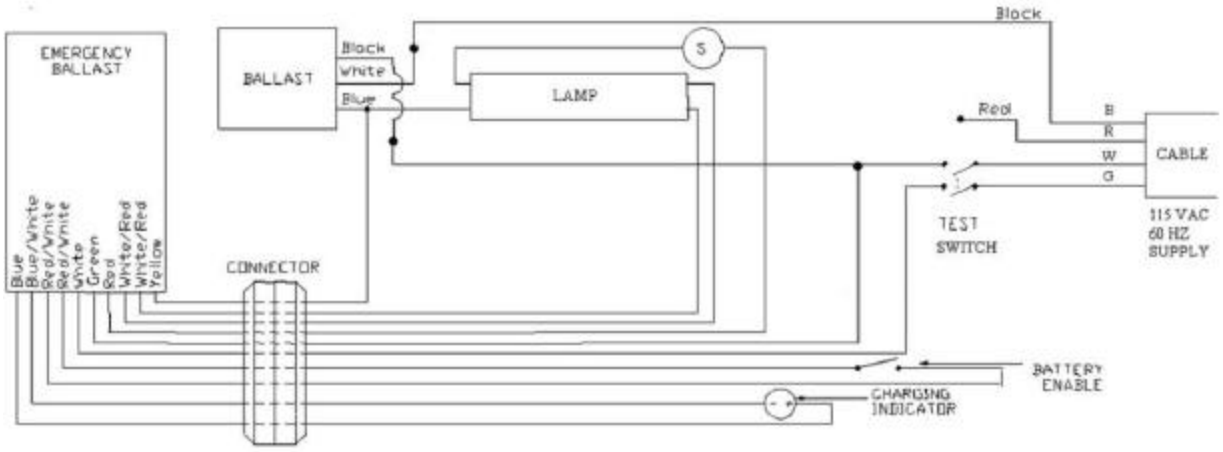


NOTES:

1. For dimensional data, see Figures 1, 2, and 3 of MIL-DTL-16377/8.
2. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

FIGURE 2. Typical view of symbols 331.1EM, 331.2EM, 347.2EM and 347.3EM.

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NOTES:

1. Ballast black & white wiring not color coded with input cable black & white, emergency lamp only.
2. Emergency ballast white & green wiring not color coded with input cable white & green.

FIGURE 3. Wiring diagram

REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

331.1EM: Complete lighting fixture assembly, standard mounting, with one 20-watt cool white fluorescent lamp, white diffusing window assembly and fluorescent emergency ballast.
331.2EM: Complete lighting fixture assembly, flush mounting, with one 20-watt cool white fluorescent lamp, white diffusing window assembly and fluorescent emergency ballast.
347.2EM: Complete lighting fixture assembly, standard mounting, with one 20-watt cool white fluorescent lamp, clear and white prismatic window assembly, and fluorescent emergency ballast.
347.3EM: Complete lighting fixture assembly, flush mounting, with one 20-watt cool white fluorescent lamp, clear and white prismatic window assembly, and fluorescent emergency ballast.

DESIGN:

Electrical rating: 20watts, 120 volts, 60 hertz.
Dimension and configurations: See Figures 1, 2, and 3 of MIL-DTL-16377/8 and Figure 10 of MIL-DTL-16377.
Photometric data: See Figure 4 of MIL-DTL-16377/8.
Lampholder spacing: See Figure 21 of MIL-DTL-16377
Wiring diagram: See Figure 3.
Mounting dimensions:
Symbols 331.1EM and 347.2EM: See Figure 25 of MIL-DTL-16377.
Symbols 331.2EM and 347.3EM: See Figure 26 of MIL-DTL-16377.
Enclosure: Watertight. See 3.5.11 of MIL-DTL-16377.
Cable: 10 feet of type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Gasket: Neoprene rubber (durometer 35-45)
Weight:
Symbols 331.1EM and 347.2EM: 13.5 lbs. maximum
Symbols 331.2EM and 347.3EM: 15 lbs. maximum

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Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a), (b), and (c) in accordance with MIL-DTL-16377/8. Labels (d), (e), and (f) as shown on Figure 2:

(a) Two type I labels.
(b) One type II label for Symbols 331.2EM and 347.3EM only.
(c) One type IV label. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY".
(d) One type IV label. The information to be shown on the type IV label shall be "CHARGING INDICATOR LIGHT".
(e) One type IV label. The information to be shown on the type IV label shall be "BATTERY ENABLE".
(f) One type V label. The information to be shown on the type V label shall be "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION".

Standard Parts: The standard parts and their respective part numbers shown in Table I shall be used in the construction of the fixtures specified herein.

TABLE I. Standard parts.

Part Number	Description	Specification/Material	Remarks
M16377/73-001	Lighting assembly	M16377/8-331.1	For use with Symbol 331.1EM
M16377/73-002	Lighting assembly	M16377/8-331.2	For use with Symbol 331.2EM
M16377/73-003	Lighting assembly	M16377/8-347.2	For use with Symbol 347.2EM
M16377/73-004	Lighting assembly	M16377/8-347.3	For use with Symbol 347.3EM
M16377/73-005	Mating harness	AMP part No. 206043-1 or equal	Mates with M16377/77-001
M16377/73-006	Indicator light	Commercial	LED type
M16377/73-007	Switch (DPST)	MS16569	
M16377/73-008	Enclosure for emergency ballast	Commercial	Aluminum; Approximately 3" H x 3½" W x 16" L
M16377/73-009	Test Switch	Commercial	Double pole, Test Switch with rubber boot
M16377/77-001	Emergency ballast and wiring harness	AMP part No. 206043 or equal	Mates with M16377/73-005

NOTE: Switch shall normally be left in the "Battery Enable" position.

EXAMINATION AND TESTS: Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See Table II.
Conformance:
Comparison inspection: Same as the first article inspection.

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TABLE II. First article and comparison inspection for complete fixtures.

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	Candlepower
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

Custodians:
Navy - SH
DLA - IS

Preparing activity:
Navy - SH
(Project 6210-N001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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